

Abdominal oesophagus and stomach

ABDOMINAL OESOPHAGUS

The abdominal oesophagus is 1–2.5 cm in length, and is slightly broader at the cardiac orifice than at the diaphragmatic aperture. It lies to the left of the midline and enters the abdomen through the oesophageal aperture (formed by the two diaphragmatic crura) at the level of the eleventh thoracic vertebra (Mirjalili et al 2012). It runs obliquely to the left and slightly posteriorly, and ends at the gastro-oesophageal junction, where it is continuous with the cardiac orifice of the stomach. The anterior wall of the abdominal oesophagus is effectively longer than the posterior wall because of the obliquity of the crura. The abdominal oesophagus lies posterior to the left lobe of the liver, and anterior to the left crus, the left inferior phrenic vessels and the left greater and lesser splanchnic nerves; its surface is covered by a thin layer of connective tissue and visceral peritoneum that contain the anterior and posterior vagus nerves, as well as the oesophageal branches of the left gastric vessels. The anterior and posterior vagi may be single or composed of multiple trunks (Jackson 1949); the anterior is closely applied to the anterior outer surface of the longitudinal muscle coat of the oesophagus while the posterior usually lies within loose connective tissue immediately posterior and to the right of the oesophagus, making its identification during surgery somewhat easier. Unlike the more proximal parts of the oesophagus, the muscular wall of the abdominal oesophagus is composed entirely of smooth muscle.

The abdominal oesophagus is effectively tethered to the margins of the muscular oesophageal hiatus in the diaphragm by the phreno-oesophageal ligament (Fig. 64.1) (Kwok et al 1999). This is formed by two circumferential layers of elastin-rich connective tissue containing some smooth muscle fibres. The inferior layer is an extension of the subperitoneal transversalis fascia below the diaphragm; it is thin and only loosely attached to the oesophagus. The superior layer is continu-

ous with the subpleural endothoracic fascia above the diaphragm; it is thicker, contains more elastin than its inferior counterpart, and runs cranially and obliquely to fuse firmly with the wall of the oesophagus. A variable amount of adipose tissue lies in the triangular interval between the two layers of the ligament. The phreno-oesophageal ligament helps to anchor the oesophagus to the crural muscle fibres of the diaphragm and probably acts to limit upward and downward mobility of the oesophagus within the hiatus (Kwok et al 1999). In the elderly, the ligament tends to become attenuated and contain more adipose tissue. The phreno-oesophageal ligament is denser anteriorly where it bridges between the outer layer of the oesophageal wall and the arching fibres of the diaphragmatic crura.

The peritoneal reflection posterior to the abdominal oesophagus is short and continues directly on to the posterior surface of the fundus of the stomach; it is sometimes referred to as the gastrophrenic ligament. It encloses the oesophageal branches of the left gastric vessels and the coeliac branches of the posterior vagus and can thus be considered to form an extremely short, wide mesentery to the abdominal oesophagus. In adults, a fat pad may be visible beneath the peritoneum over the anterior surface of the gastro-oesophageal junction and can be a useful surgical marker of the gastro-oesophageal junction.

Hiatus hernia



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VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Arteries

The abdominal oesophagus is supplied by numerous oesophageal branches of the left gastric artery. These ascend beneath the visceral peritoneum to supply perforating branches to the intramural and submucosal plexuses. The posterior surface usually receives an additional supply via branches of the upper short gastric arteries, reinforced by terminal arteries from the oesophageal branches of the thoracic aorta and occasionally an ascending branch of the posterior gastric artery (Liebermann-Meffert et al 1987).

Veins

Mucosal and submucosal veins drain via plexuses to the left gastric and upper short gastric veins in the abdomen and to the azygos/hemiazygos system of veins in the thorax. The distal oesophagus is an important site of portosystemic anastomosis where oesophageal varices develop in portal hypertension (see below).

Lymphatic drainage

The oesophagus has a freely anastomosing plexus of lymphatics in the lamina propria, submucosa and muscularis propria. The lower third primarily drains caudally to left gastric and left and right paracardial nodes, and from there to coeliac nodes. In lower oesophageal cancer, lymph also drains cranially to mediastinal lymph nodes (Aikou et al 1987).

INNERVATION

The oesophagus has a well-developed intrinsic nervous system consisting of a ganglionated myenteric plexus and a sparsely ganglionated submucosal plexus, modulated by extrinsic autonomic nerves. Parasympathetic innervation of the abdominal oesophagus is derived directly

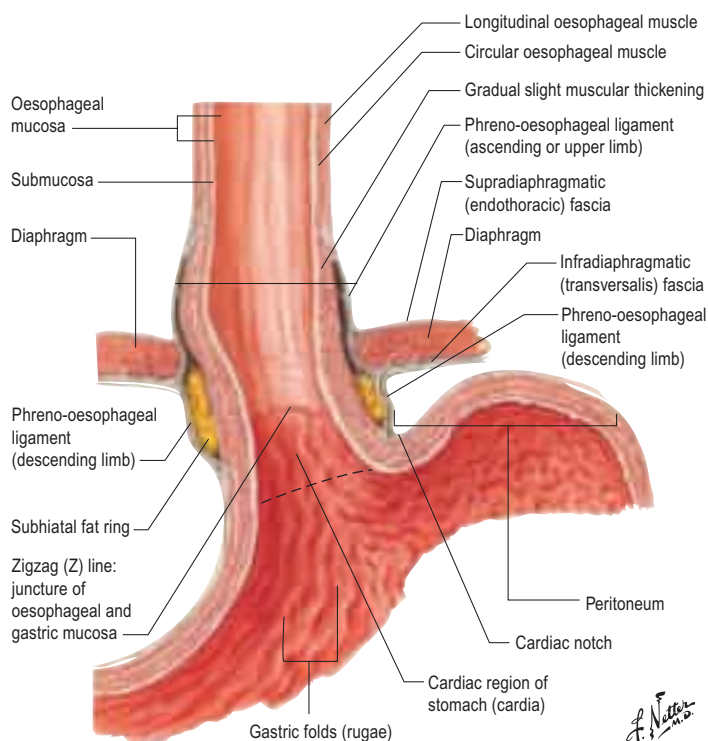


Fig. 64.1 The anatomical structures around the abdominal oesophagus. (Reprinted from Netter Anatomy Illustration Collection, © Elsevier Inc. All Rights Reserved.)

Hiatus hernia is a relatively common condition involving expansion of the oesophageal hiatus and herniation of the stomach through the diaphragm into the mediastinum. It is more common in the elderly and the obese. There are two types of hiatus hernia: the sliding type, which accounts for at least 90%, and the para-oesophageal or mixed type, which makes up the remainder ([Roman and Kahrilas 2014](#)). A hiatus hernia may not cause any symptoms or it may be associated with symptoms of gastro-oesophageal reflux. Para-oesophageal hernias can cause obstruction and/or ischaemia of the herniated stomach. Treatment of a symptomatic sliding hiatal hernia is directed at managing associated gastro-oesophageal reflux, which may require anti-reflux surgery.

from the thoracic peri-oesophageal plexus and, to a lesser extent, from the anterior and posterior vagi. These nerves are motor to the distal oesophagus and both stimulatory and inhibitory to the lower oesophageal sphincter, maintaining basal tone and coordinating distal oesophageal peristalsis with relaxation of the sphincter during swallowing (the latter being mediated by intrinsic nitrenergic inhibitory neurones under vagal control). Sympathetic supply of the distal oesophagus originates from the fifth to twelfth thoracic spinal segments mainly via the greater and lesser splanchnic nerves and the coeliac plexus. Nociceptive signals are conveyed by afferent nerves accompanying sympathetic nerves and by vagal afferents, which are also involved in mechanosensory signalling (Neuhuber et al 2006).

STOMACH

The stomach is the widest part of the alimentary tract and lies between the oesophagus and the duodenum. It performs numerous functions, including the temporary storage of ingested nutrients; mechanical breakdown of solid food; chemical digestion of proteins; regulation of the passage of chyme into the duodenum; secretion of intrinsic factor for vitamin B₁₂ absorption; secretion of gut hormones; and secretion of acid to aid digestion (including the absorption of iron). It is also important in microbial defence.

The stomach is situated in the upper abdomen, extending from the left upper quadrant downwards, forwards and to the right, lying in the left hypochondrium, epigastrium and umbilical regions. It occupies a recess beneath the diaphragm and anterior abdominal wall bounded by the upper abdominal viscera on either side. The mean capacity of the stomach increases from approximately 20–30 ml at birth to approximately 1000–1500 ml in adults. The peritoneal surface of the stomach is interrupted by the attachments of the greater and lesser omenta, which define the greater and lesser curvatures and separate the anterior and posterior surfaces.

PARTS OF THE STOMACH

For descriptive purposes, the stomach can be divided into a fundus, body, pyloric antrum and pylorus by artificial lines drawn on its external surface (Fig. 64.2). The internal appearance and microstructure of these regions vary. The fundus is dome-shaped and projects above and to the left of the oesophageal opening (cardiac (cardial) orifice) to lie in contact with the left dome of the diaphragm; it lies above a horizontal line from the cardiac notch to the greater curvature. The body extends from the fundus to the angular incisure (incisura angularis), a constant external notch at the lower end of the lesser curvature. The cardia is the region of the stomach adjacent to the oesophageal opening. A line

drawn from the angular incisure to an inconstant indentation on the greater curvature defines the lower boundary of the body. The pyloric antrum extends from this line to where the stomach narrows to become the pyloric canal (1–2 cm long), which terminates at the pyloric orifice (Didio and Anderson 1968).

GASTRIC RELATIONS

Gastric curvatures

Lesser curvature

The lesser curvature extends between the cardiac and pyloric orifices and forms the medial border of the stomach. It descends from the medial side of the oesophagus in front of the decussating fibres of the right crus of the diaphragm, curves downwards and to the right, and lies anterior to the superior border of the pancreas (Fig. 64.3). It ends at the pylorus, just to the right of the midline. In the most dependent

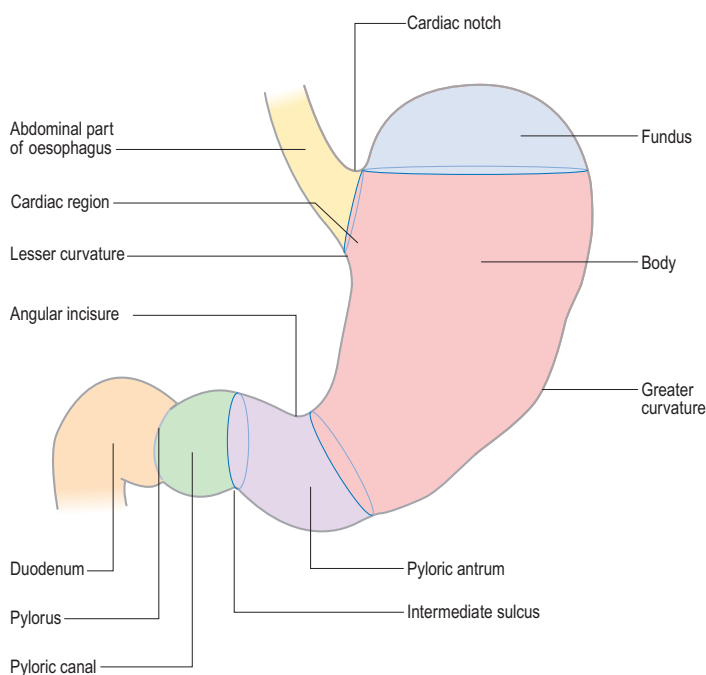


Fig. 64.2 The parts of the stomach.

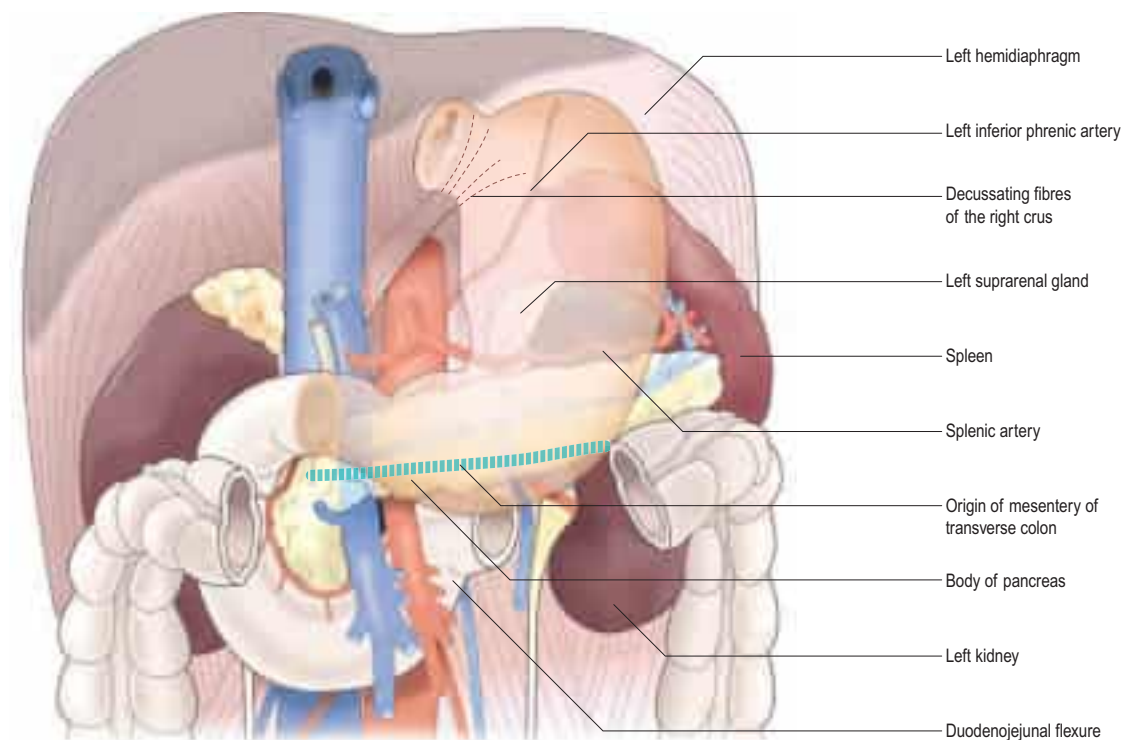


Fig. 64.3 The posterior relations of the stomach.

part, there is typically a notch, the angular incisure, whose position and appearance vary with gastric distension. The lesser omentum is attached to the lesser curvature and contains the right and left gastric vessels.

Greater curvature

The greater curvature is two to three times longer than the lesser (Csendes and Burgos 2005). It starts from the cardiac notch, formed between the lateral border of the abdominal oesophagus and the fundus of the stomach, and arches upwards, posterolaterally and to the left. Its highest convexity, the apex of the fundus, is approximately level with the left sixth rib anteriorly but varies between individuals and with respiration (Mirjalili et al 2012). From this point, it sweeps inferiorly and anteriorly, slightly convex to the left, almost as far as the tenth costal cartilage in the supine position, where it turns medially to end at the pylorus in the transpyloric plane at the lower border of the first lumbar vertebra. There is frequently a groove, the intermediate sulcus, in the curvature close to the pyloric canal. The start of the greater curvature is covered by peritoneum, which continues over the anterior surface of the stomach. Laterally, the greater curvature gives attachment to the gastrosplenic ligament and, below this, to the greater omentum, which contains the gastroepiploic vessels. The gastrosplenic ligament and the greater omentum, together with the gastrophrenic and spleno-renal ligaments, are continuous derivatives of the original dorsal mesogastrium; their names merely indicate regions of the same continuous sheet of peritoneum and its associated connective tissue.

Gastric volvulus



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Gastric surfaces

When the stomach is empty and contracted, the anterior and posterior surfaces tend to face superiorly and inferiorly, but as the stomach distends, they face progressively more anteriorly and posteriorly (Fig. 64.4).

Anterior (superior) surface

The entire anterior (superior) surface of the stomach is covered by peritoneum. The lateral part of the anterior surface lies posterior to the left costal margin in contact with the diaphragm, which separates it from the left pleura, the base of the left lung, the pericardium and the left seventh to ninth ribs and costal cartilages (Fig. 64.5). It also lies posterior to the costal attachments of the upper fibres of transversus abdominis. The upper left part of this surface curves posteriorly and lies in contact with the visceral surface of the spleen. The right half of the anterior surface is related superiorly to the left lobe of the liver and inferiorly to the anterior abdominal wall, through which it can be accessed by a needle for placement of a gastrostomy tube (see below). When the stomach is empty, the transverse colon may lie adjacent to the anterior surface.

Posterior (inferior) surface

The posterior surface is covered by peritoneum, except near the cardiac orifice, where a small, triangular area contacts the left diaphragmatic

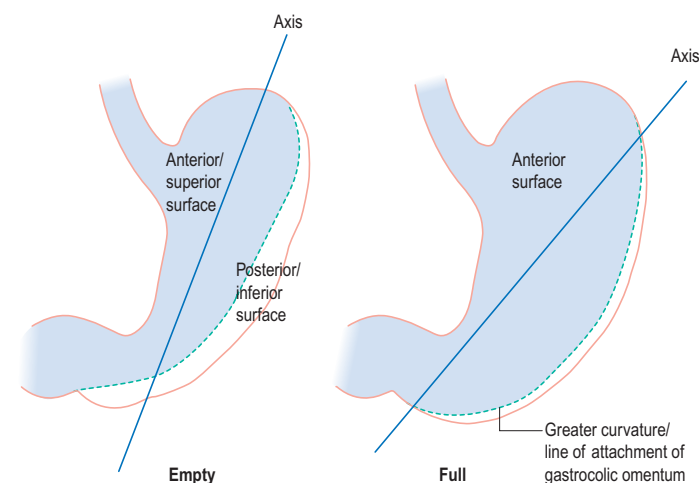


Fig. 64.4 Axes of the empty and full stomach. As the stomach distends, the greater curvature 'rolls' downwards and the anterosuperior surface comes to lie almost completely vertical as the anterior surface.

crus and sometimes the left suprarenal gland. The left gastric vessels reach the lesser curvature at the right extremity of this bare area. The gastrophrenic ligament passes from the lateral aspect of this bare area to the inferior surface of the diaphragm. The posterior surface of the stomach lies anterior to the left crus and lower fibres of the diaphragm, the left inferior phrenic vessels, the left suprarenal gland, the superior pole of the left kidney, the splenic artery, the anterior surface of the pancreas, and the upper layer of the transverse mesocolon (see Fig. 64.3). Together, these form the shallow stomach bed, which is separated from the stomach by the lesser sac (over which the stomach slides as it distends). The upper left part of the posterior surface curves anterolaterally and lies in contact with the visceral surface of the spleen. The transverse mesocolon separates the stomach from the duodenojejunal flexure and proximal jejunum.

GASTRIC ORIFICES

Cardiac orifice and gastro-oesophageal junction

The oesophagus opens into the stomach at the cardiac orifice (see Fig. 64.1). It is usually situated to the left of the midline behind the seventh costal cartilage at the level of the eleventh thoracic vertebra (Mirjalili et al 2012). It is, on average, about 40 cm from the incisor teeth in the adult. The right side of the abdominal oesophagus is continuous with the lesser curvature and the left side with the greater curvature.

Internally, the transition between the oesophagus and stomach is difficult to define because gastric fundal mucosal folds extend a variable distance up the abdominal oesophagus. For practical purposes, the gastro-oesophageal junction is usually identified by a circumferential 'zigzag' line ('Z line') between the pale pink oesophageal squamous epithelium above and the red columnar epithelium below. In the healthy oesophagus, the Z line is located at the proximal extent of the gastric mucosal folds. When metaplastic columnar epithelium extends above the gastric folds into the lower oesophagus, it is regarded as pathological and termed Barrett's oesophagus. The presence of a sliding hiatus hernia with or without Barrett's oesophagus can make

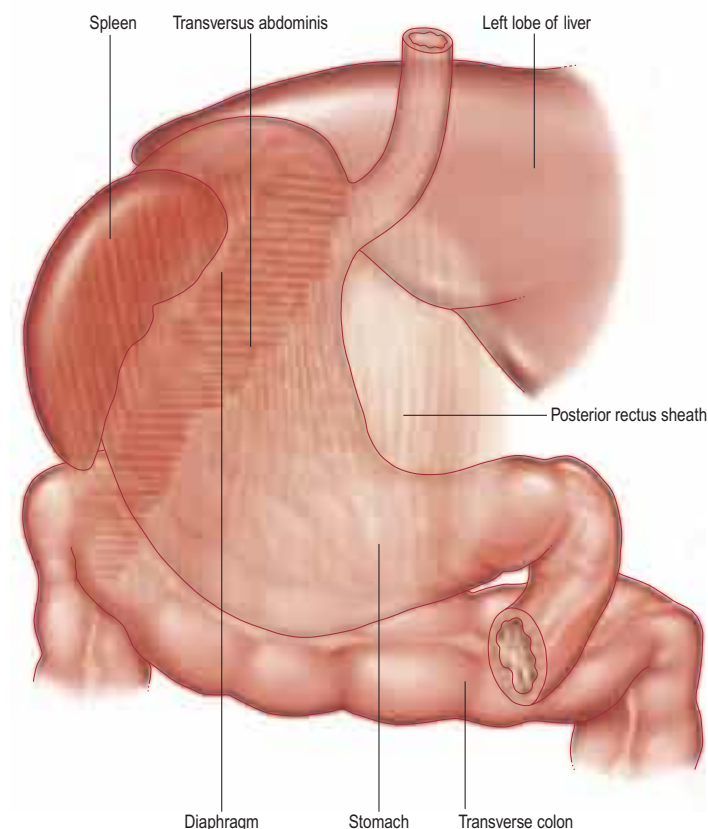


Fig. 64.5 Anterior relations of the stomach, viewed from behind. The fibres of the diaphragm (oblique), transversus abdominis (transverse) and rectus abdominis (vertical) can be seen through the outlines of the stomach and its visceral relations.

Volvulus of the stomach is much less common than volvulus of either the sigmoid colon or the caecum. Two types of gastric volvulus may occur. The first, organoaxial volvulus, occurs about an axis of rotation running from below the cardiac orifice to the pylorus. The antrum, body and fundus rotate upwards, such that the greater curvature comes to lie above the lesser curvature. The second type, mesenteroaxial volvulus, occurs about an axis drawn 'across' the body of the stomach, usually just above the angular incisure. This type of volvulus is perpendicular to the axis of organoaxial volvulus. The distal body and antrum rotate anteriorly, superiorly and laterally while the upper body and fundus rotate posteriorly, medially and inferiorly. Although it is relatively mobile within the upper abdomen, the stomach is normally tethered at the gastro-oesophageal junction and the pylorus, to the spleen by the gastrosplenic ligament, and to the liver by the lesser omentum. The attachment to the transverse colon via the gastrocolic omentum also restrains the stomach but it is the most mobile attachment. For either type of gastric volvulus to occur, some or all of these points of tethering must be loosened either by previous surgical division or by chronic lengthening and laxity of their connective tissue. Organoaxial volvulus is most common because the lesser omentum, gastrosplenic ligament and gastrocolic omentum are more likely to undergo chronic lengthening by traction than the other attachments of the stomach. Mesenteroaxial volvulus requires the gastro-oesophageal junction and pylorus to be sufficiently mobile as to come into close approximation, which is much less common. Despite the rich arterial supply of the stomach, either type of volvulus may cause a strangulating obstruction.

Provided the detorted stomach is viable, gastric volvulus is typically managed by fixation of the anterior surface of the stomach to the anterior abdominal wall ('gastropexy') ([García et al 2013](#)). Gastric volvulus of the organoaxial type is frequently associated with a para-oesophageal hiatus hernia, which must also be repaired. This surgery can usually be carried out laparoscopically.

endoscopic identification of the gastro-oesophageal junction particularly difficult.

Externally, a fat pad is often present in adults beneath the peritoneum over the anterior surface of the gastro-oesophageal junction. The junction is also demarcated by a sling of longitudinal and oblique gastric muscle fibres that forms a loop on the superior, left, side of the gastro-oesophageal junction between the oesophagus and the stomach.

Lower oesophageal sphincter and gastro-oesophageal reflux

At rest, there is a pressure gradient across the gastro-oesophageal junction, reflecting the negative intrathoracic pressure transmitted to the thoracic oesophagus and the positive intra-abdominal pressure transmitted to the stomach (augmented by any contraction of the stomach). Several anatomical and physiological factors normally prevent gastro-oesophageal reflux. The two major anti-reflux mechanisms are the tonically contracted, specialized, thickened intrinsic circular smooth muscle of the lower oesophagus, which is reinforced by the extrinsic encircling fibres of the right diaphragmatic crus. Together, they exert a radial pressure on the lower 2–4 cm of oesophagus, creating an effective high-pressure zone (HPZ) that can be measured by electromyography or manometry (Paterson 2001, Mittal and Goyal 2006, Hershovici et al 2011). The description of the intrinsic muscles of the lower oesophageal sphincter can be refined further to include clasp-like semicircular smooth muscle fibres on the right side of the oesophagus and sling-like oblique gastric muscle fibres on the left side (Preiksaitis and Diamant 1997, Stein et al 1995). The distribution of these muscle fibres corresponds to the anatomical and functional asymmetry of the HPZ.

At and just below the level of entry of the abdominal oesophagus into the stomach, the circular fibres of the intermediate layer of the muscularis externa lying over the upper lesser curvature are particularly pronounced; they are sometimes referred to as 'clasp' fibres and exert fairly constant myogenic tone (Fig. 64.6). Since the oesophagus passes obliquely into the stomach, the tone in the clasp fibres rises with increasing gastric distension and they may act to draw the anterior and posterior surfaces together, increasing the tone at the gastro-oesophageal junction and contributing to the HPZ. Several minor anti-reflux factors also exist: the folds of gastric mucosa present at the gastro-oesophageal junction form a mucosal rosette that helps to create a fluid- and gas-tight seal from the basal tonic contraction of the muscular wall of the lower oesophagus; the angle of the cardiac orifice (angle of His), formed, in part, by the pull of the oblique fibres of the innermost layer of gastric smooth muscle, thereby constituting a type of 'flap valve'; and the length of the abdominal oesophagus, which is buttressed externally by pads of adipose tissue at and below the level of the oesophageal hiatus. These anatomical and physiological features are collectively referred to as the lower oesophageal sphincter. The oesophageal part of the lower oesophageal sphincter is controlled by the intramural plexuses of the enteric nervous system; activation of inhibitory neurones

releasing nitric oxide leads to relaxation of the sphincter. Muscle tone is reduced in advance of the oesophageal peristaltic wave during swallowing and raised again after the food bolus has passed. During inspiration, the greater negative intrathoracic pressure increases the gastro-oesophageal pressure gradient, but this is offset by increased pressure in the HPZ from contraction of the peri-oesophageal fibres of the right diaphragmatic crus. Activation of the crural muscle slightly before the rest of the diaphragmatic muscle facilitates this anti-reflux mechanism.

Reflux of gastric contents into the lower oesophagus as a result of transient relaxation of the lower oesophageal sphincter occurs as a normal event in most individuals on a daily basis. Prolonged episodes of acid reflux as a result of a weak lower oesophageal sphincter or sliding hiatus hernia may cause oesophagitis and its associated complications. Surgical procedures to prevent abnormal reflux aim to restore a normal length of intra-abdominal oesophagus, repair any hiatus hernia and increase the pressure surrounding the intra-abdominal oesophagus; this is usually achieved by wrapping the fundus of the stomach around the intra-abdominal oesophagus (fundoplication).

The converse type of lower oesophageal sphincter dysfunction occurs in achalasia, which is characterized by reduced or absent ganglion cells in the myenteric plexus of the distal oesophagus and gastro-oesophageal junction. Loss of inhibitory innervation of the sphincter leads to impaired relaxation, causing dysphagia and proximal oesophageal dilation. Management of this condition involves disruption of the hypertonic muscle fibres by balloon dilation or surgical myotomy ('Heller's cardiomyotomy').

Barrett's oesophagus

The squamous epithelium lining the lower oesophagus may be pathologically replaced by a metaplastic 'intestinalized' columnar epithelium known as 'Barrett's oesophagus' (see Commentary 1.4). This process results from chronic episodic reflux of gastric acid and bile into the lower oesophagus (Dvorak et al 2007). The abnormal intestinal metaplasia may extend proximally for a variable length from the gastro-oesophageal junction and associated macroscopic mucosal changes can be seen on endoscopy. The presence of Barrett's oesophagus confers a significantly increased risk of oesophageal adenocarcinoma, although the condition can remain stable or even regress with appropriate medical therapy (Yousef et al 2008).

Pyloric orifice

The pyloric orifice is the opening from the stomach into the duodenum. When the body is supine and the stomach empty, it typically lies 1–2 cm to the right of the midline in the transpyloric plane (the lower border of the body of the first lumbar vertebra). The pyloric sphincter is formed by a circumferential thickening of circular muscle interlaced with connective tissue septa and some longitudinal muscle fibres, and is palpably thicker than the adjacent stomach and duodenum. The circular pyloric constriction on the surface of the stomach usually indicates the location of the pyloric sphincter, and may be marked by a prepyloric vein that crosses the anterior surface in a caudal direction.

Infantile hypertrophic pyloric stenosis is due to idiopathic hypertrophy of the circular muscle of the pylorus and causes gastric outlet obstruction in early infancy. It is readily treated by dividing the thickened pyloric muscle (pyloromyotomy).

GASTRIC FORM AND INTERNAL APPEARANCES

Numerous factors influence both the form and the position of the stomach, including the posture and build of the individual, the extent to which the stomach is filled, the position of the surrounding viscera, and the tone of the abdominal wall and gastric musculature. The empty stomach is most commonly J-shaped, the fundus usually contains gas, and, in the erect posture, the pylorus descends to the level of the second or third lumbar vertebra. The lowest part of the antrum may lie below the level of the umbilicus, and the overall axis of the organ is slightly inclined from the vertical (Fig. 64.7). In short, obese individuals, the axis of the stomach lies more towards the horizontal as a 'steer-horn' shape.

The extent of filling mainly affects the shape and position of the body of the stomach. As the stomach fills, it expands forwards and downwards but, when the colon or small bowel is distended, the fundus enlarges towards the liver and diaphragm. As stomach capacity increases, the pylorus is displaced to the right and the axis of the whole organ lies in a more oblique direction (see Fig. 64.4).

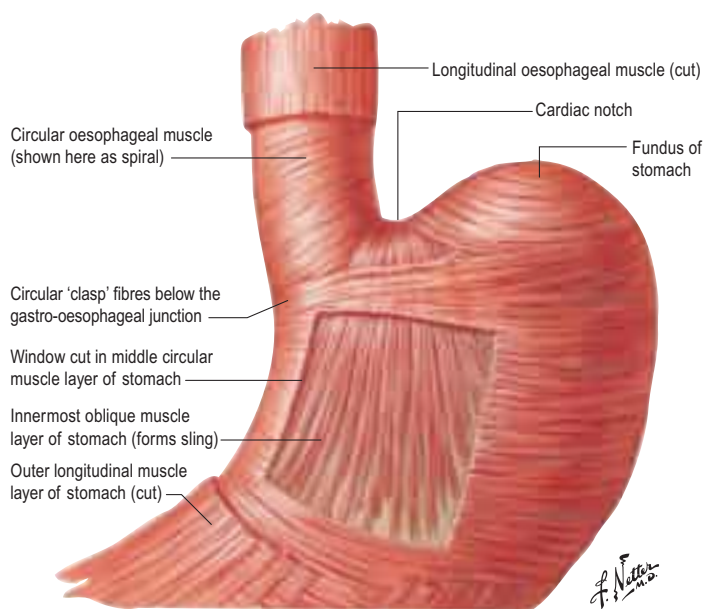


Fig. 64.6 Detail of the arrangements of the muscle layers of the oesophageal and gastric walls. (Reproduced and modified from Netter Anatomy Illustration Collection, © Elsevier Inc. All Rights Reserved.)

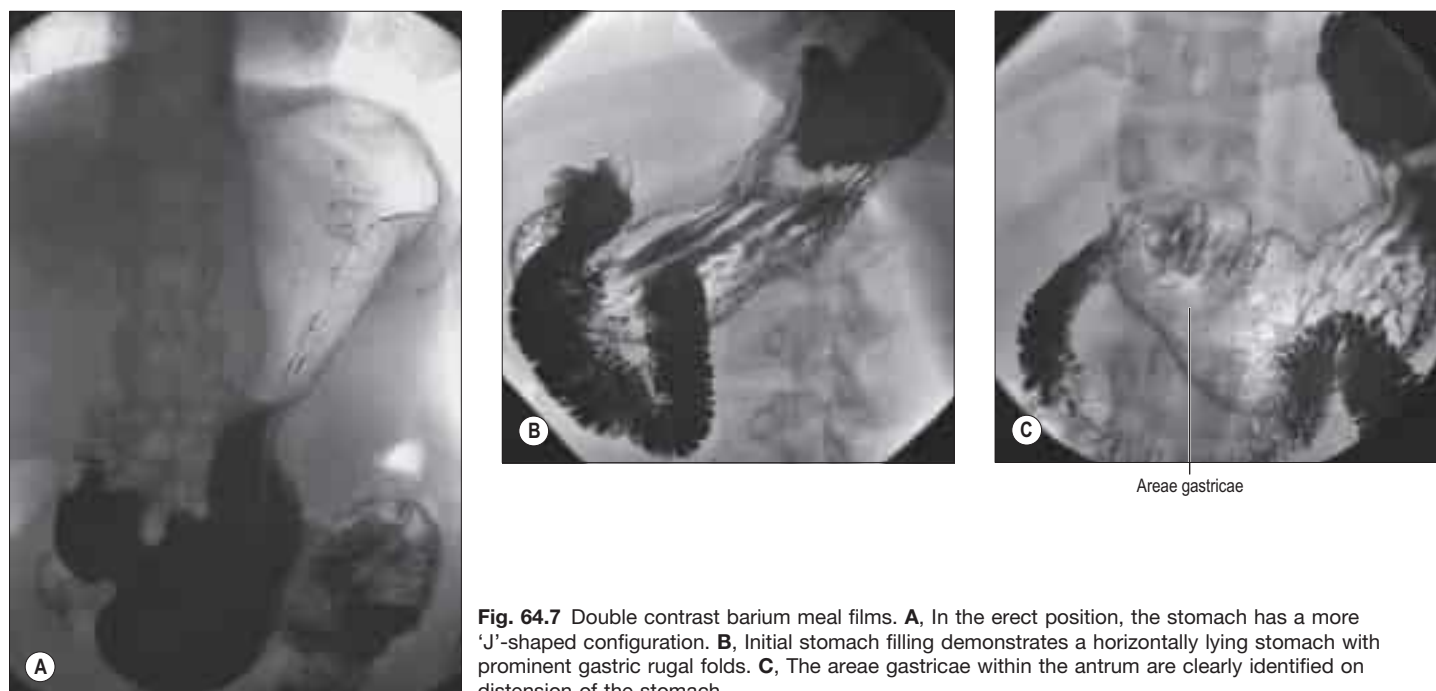


Fig. 64.7 Double contrast barium meal films. **A**, In the erect position, the stomach has a more 'J'-shaped configuration. **B**, Initial stomach filling demonstrates a horizontally lying stomach with prominent gastric rugal folds. **C**, The areae gastricae within the antrum are clearly identified on distension of the stomach.

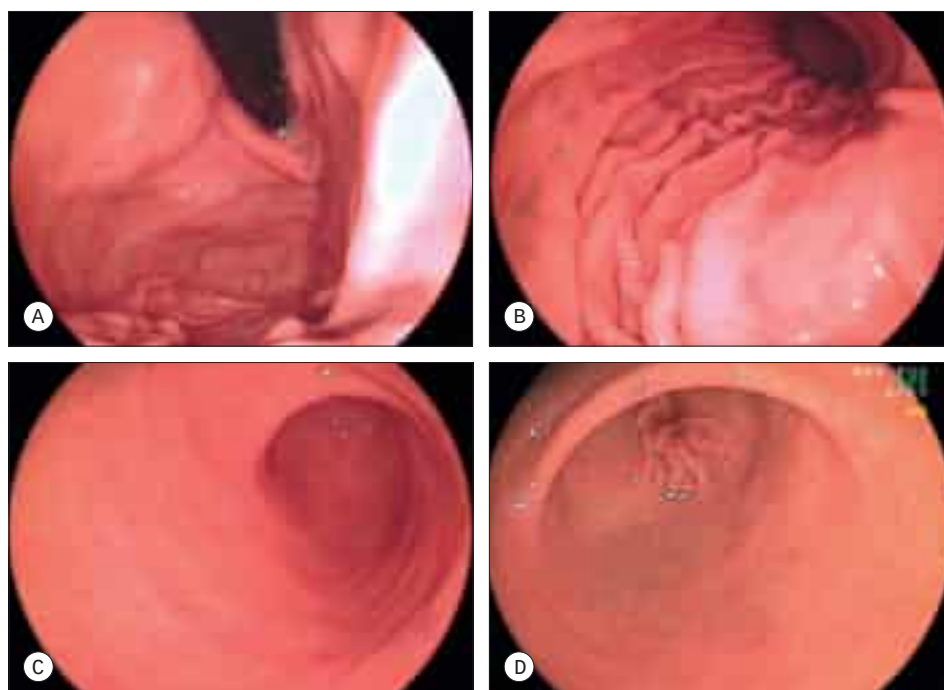


Fig. 64.8 Endoscopic appearances of the stomach. **A**, The cardiac orifice and fundus from below, showing the pronounced lateral mucosal fold at the cardiac orifice. **B**, The body and greater curvature, showing mucosal folds ('magenstrasse'). **C**, The antrum, showing the internal appearance of the incisura angularis. **D**, The prepyloric antrum and opening of the pyloric canal.

Internal appearances

During endoscopic examination (Fig. 64.8), the stomach is typically at least partially distended by air. The cardiac orifice and the lowest portion of the abdominal oesophagus viewed from above are typically closed at rest by tonic contraction of the lower oesophageal sphincter. The transition between the pale pink squamous epithelium of the oesophagus and the red columnar epithelium of the stomach, the 'Z line', is usually clearly visible at the proximal extent of the gastric mucosal folds (Silverstein and Tytgat 1991). Viewed retrogradely from within the distended stomach, the cardiac orifice lies medial to the fundus, with a mucosal fold between the two corresponding to the acute angle at this orifice. The mucosa also appears slightly thickened, forming part of the 'mucosal rosette' that lines the orifice and contributes to its closure (see above).

In the partially distended stomach, the mucosa of the body and, to a lesser extent, the fundus is thrown into longitudinal folds (rugae) (see Fig. 64.7). These are most obvious on the anterolateral, lateral and posterolateral parts of the stomach, towards the greater curvature. The smoother mucosa along the lesser curvature forms a 'gastric canal' or 'magenstrasse' that enables liquids entering the stomach to be fast-tracked to the pylorus (see Fig. 64.8).

The areae gastricae within the antrum are small undulations of the mucosal surface that give a subtle cobblestone-type appearance on double contrast barium meal examination (see Fig. 64.7) (Mackintosh and Kreel 1977). The few folds present in the antrum when the stomach is relaxed disappear with distension. The antrum adjacent to the pyloric canal, the prepyloric antrum, has a smooth mucosal surface that culminates in a slight puckering of the mucosa at the pyloric orifice, caused by contraction of the pyloric sphincter.

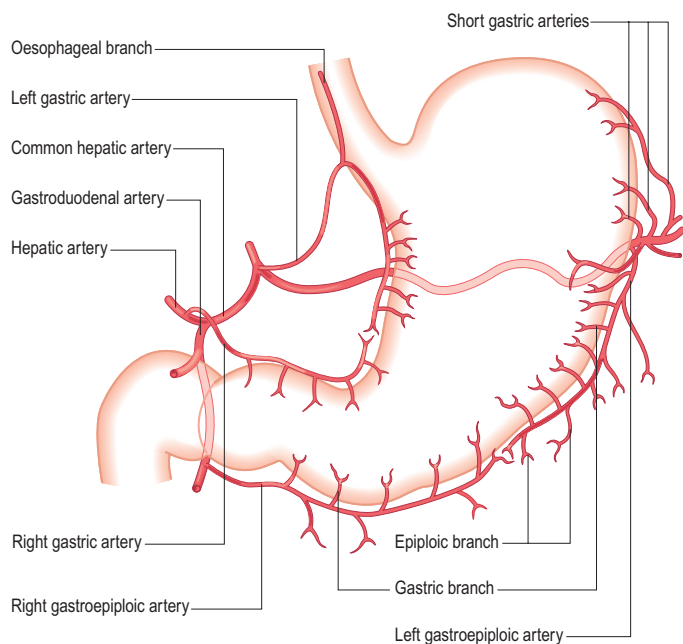


Fig. 64.9 The arterial supply of the stomach. A posterior gastric artery is not shown in this illustration.

Gastrectomy



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VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Arteries

The arterial supply to the stomach comes predominantly from the coeliac trunk, although intramural anastomoses exist with arteries of other origins at the two ends of the stomach (Figs 64.9–64.11). The left gastric artery usually arises directly from the coeliac trunk. The short gastric arteries, left gastroepiploic artery and, when present, the posterior gastric artery are branches of the splenic artery. The right gastric artery and right gastroepiploic artery arise from the hepatic artery and its gastroduodenal branch, respectively.

Left gastric artery

The left gastric artery is the smallest branch of the coeliac trunk. It ascends to the left of the midline and crosses over the lower end of the left crus of the diaphragm beneath a fold of peritoneum in the upper posterior wall of the lesser sac (the gastropancreatic fold). Here, it lies adjacent to the left inferior phrenic artery and medial or anterior to the left suprarenal gland. It runs forwards into the superior portion of the lesser omentum adjacent to the upper end of the lesser curvature, and then turns anteroinferiorly to run along the lesser curvature between the two peritoneal leaves of the lesser omentum. At the highest point of its course, it gives off one or more oesophageal branches. In its course along the lesser curvature, it gives off multiple branches that run on to the anterior and posterior surfaces of the stomach, after which it anastomoses with the right gastric artery in the region of the angular incisure.

The left gastric artery (replaced or accessory) may rarely arise from the common hepatic artery or its left branch, or directly from the abdominal aorta (Panagouli et al 2013). The most common of these variations is an origin from the left branch of the hepatic artery, when the left gastric artery passes between the peritoneal layers of the upper lesser omentum to reach the lesser curvature of the stomach. However, a replaced/accessory left hepatic artery arising from the left gastric artery is more common than a replaced/accessory left gastric artery origin.

Short gastric arteries

For practical purposes, the short gastric arteries may be defined as those arteries arising above the level of the splenic artery and supplying the fundus of the stomach on its greater curvature. The short gastric arteries are variable in number: commonly, between five and seven. They arise from the splenic artery or its terminal divisions, or from the proximal left gastroepiploic artery, and pass between the layers of the gastrosplenic ligament to supply the gastric fundus and cardiac orifice. They

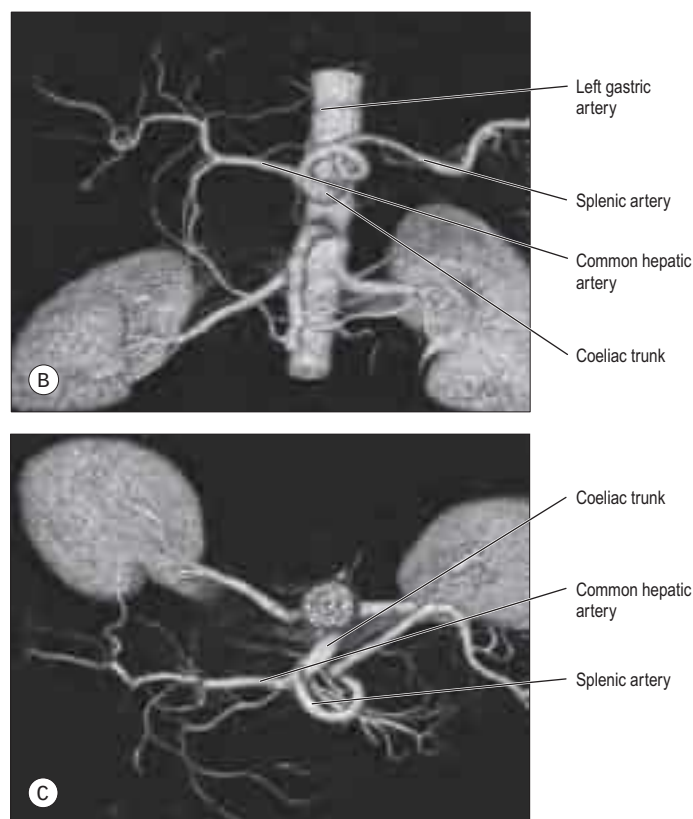
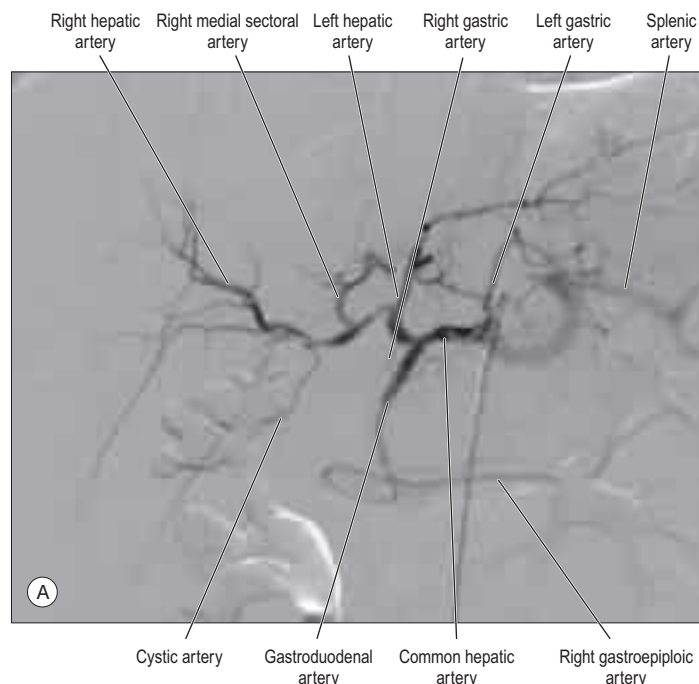


Fig. 64.10 The coeliac trunk and its branches. **A**, Digital subtraction angiogram. **B–C**, Digital reformat multislice CT angiograms. (**A**, Courtesy of Dr James McCall, Chelsea & Westminster Hospital, London. **B** and **C**, Courtesy of GE Worldwide Medical Systems.)

anastomose with branches of the left gastric and left gastroepiploic arteries. Rarely, an accessory left gastric artery may arise with these vessels from the distal splenic artery.

Left gastroepiploic artery

The left gastroepiploic artery is the largest branch of the splenic artery. It arises near the splenic hilum and runs anteroinferiorly between the layers of the gastrosplenic ligament into the upper gastrocolic omentum. Here, it descends between the layers of peritoneum close to the greater curvature and often anastomoses with the right gastroepiploic artery. It gives off gastric branches to the fundus of the stomach through the gastrosplenic ligament, and to the body of the stomach through the gastrocolic omentum. These are necessarily longer than the gastric branches of the right gastroepiploic artery and may be up to 8 cm long.

Since the lower half of the anterior surface of the stomach lies adjacent to the anterior abdominal wall in the left upper quadrant, it may be readily accessed to form a gastrostomy. The mobility of the stomach allows it to be approximated to the parietal peritoneum on the upper anterior abdominal wall, and a communication can then be established between the lumen of the stomach and the skin surface. This may be performed as an open surgical procedure but is much more commonly undertaken by percutaneous puncture of the stomach, guided by an endoscope within the gastric lumen or by radiological or laparoscopic visualization of the stomach. Caution is needed to avoid inadvertent transfixion of the transverse colon, which may be interposed between the stomach and anterior abdominal wall.

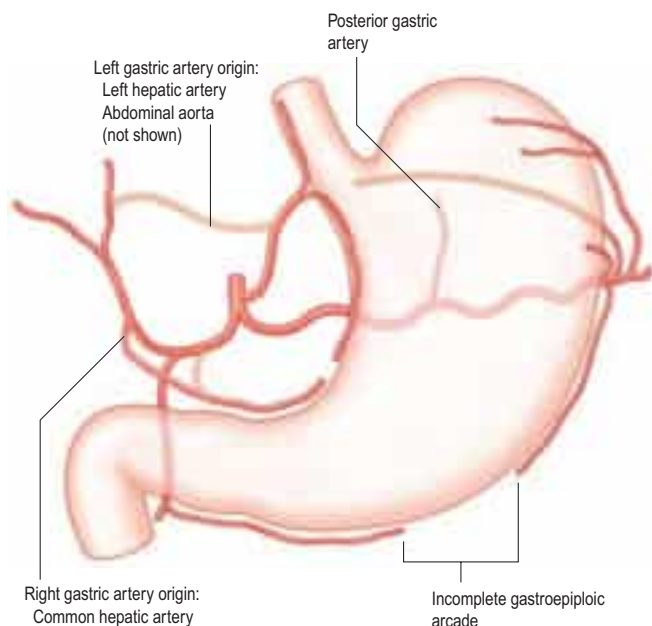


Fig. 64.11 Principal variations in the gastric arterial supply. Accessory vessels or possible replaced origins of vessels are shown by pale pink lines.

Epiploic (omental) branches arise along the course of the vessel and descend between the layers of the gastrocolic omentum into the greater omentum. A particularly large epiploic branch commonly originates close to the origin of the left gastroepiploic artery, descends in the lateral portion of the greater omentum and provides a large arterial supply to the lateral half of the omentum.

Posterior gastric artery

A posterior gastric artery supplying the posterior wall of the upper part of the gastric body is commonly present but there has been a lack of consensus regarding its origin, course and distribution. It usually arises from the splenic artery (usually from its mid section), posterior to the body of the stomach (see Fig. 64.11), and ascends behind the peritoneum of the lesser sac towards the fundus to reach the posterior surface of the stomach. It may also arise from the left gastric artery or coeliac trunk (Loukas et al 2007).

Right gastric artery

The right gastric artery is a relatively small artery that usually arises from the hepatic artery proper and runs forwards into the lesser omentum just above the first part of the duodenum. It then travels within the lesser omentum along the lesser curvature of the stomach, giving off multiple branches to the anterior and posterior surfaces of the stomach, before anastomosing with the left gastric artery. In a significant minority of individuals, it may originate from the common hepatic artery, left hepatic artery or gastroduodenal artery (Yamagami et al 2010).

Right gastroepiploic artery

The right gastroepiploic artery usually originates from the gastroduodenal artery behind the first part of the duodenum, anterior to the head of the pancreas. It passes inferiorly towards the midline just below the pylorus and then runs laterally along the greater curvature between the layers of the gastrocolic omentum about 1–2 cm from the greater curvature of the stomach. It ends by anastomosing with the left gastroepiploic artery (although this anastomosis is variably developed (Ndoye et al 2006) and may be absent). The right gastroepiploic artery gives off gastric branches that ascend on to the anterior and posterior surfaces of the antrum and lower body of the stomach; epiploic branches that descend into the greater omentum; and branches that contribute to the supply of the inferior aspect of the first part of the duodenum.

Arterial anastomoses of the stomach

Oesophageal arteries originating from the thoracic aorta anastomose with vessels supplying the fundus of the stomach in the region of the cardiac orifice. At the pyloric orifice, the extensive network of vessels supplying the duodenum allows for some anastomosis between branches of the superior mesenteric artery and pyloric vessels derived from arteries arising from the coeliac trunk. The major named vessels

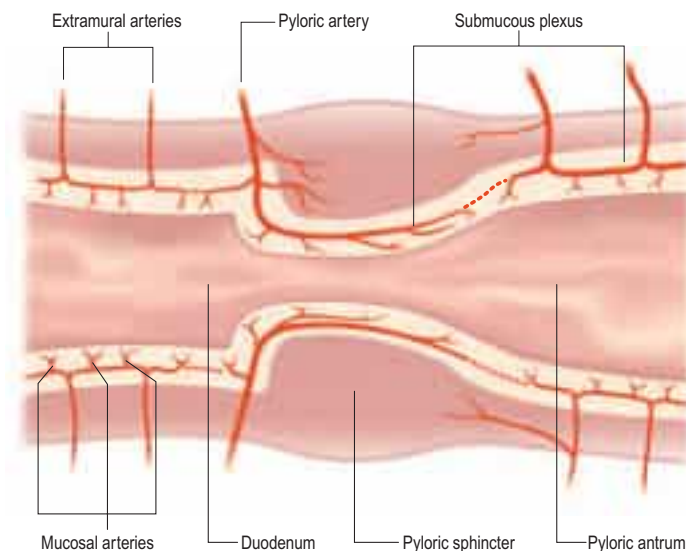


Fig. 64.12 The blood supply of the stomach and the proximal duodenum: a scheme of arterial arrangements at the gastroduodenal junction. Dotted lines indicate sites where the submucous plexus may be non-continuous. Shaded areas represent the muscular layer of the visceral wall. (Redrawn courtesy of C Piasecki, Department of Anatomy, Royal Free Hospital School of Medicine, London, and the Journal of Anatomy.)

supplying the stomach form extensive arterial anastomoses throughout the wall of the stomach but particularly in the submucosa. The right and left gastroepiploic arteries and left and right gastric arteries anastomose along the greater and lesser curvatures, respectively. Anastomoses also exist in the fundus between the short gastric and left gastric arteries, and in the antrum between the right gastric and right gastroepiploic arteries. The rich arterial supply to the stomach ensures that the high mucosal blood flow required for physiological functioning is maintained even if one or more vessels become occluded; the stomach shows considerable resistance to ischaemia, even when multiple arterial supplies are lost.

The pyloric arteries are branches of the right gastric and right gastroepiploic arteries. They pierce the duodenal wall around its entire circumference just distal to the sphincter and reach the submucosa. Here, they divide into two or three rami that run towards the stomach in the submucosa of the pyloric canal and terminate in the mucosa of the pyloric antrum (Fig. 64.12) (Piasecki 1974). The pyloric arteries anastomose with submucosal arteries in the duodenum and gastric antrum close to their origin and termination, respectively. The pyloric sphincter muscle is supplied by gastric and pyloric arteries via branches that leave their parent vessels in the subserosal and submucosal levels to penetrate the sphincter.

Dieulafoy lesions



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Veins

The veins draining the stomach ultimately empty into the portal vein. A rich submucosal and intramural venous network gives rise to veins that usually accompany the corresponding named arteries and drain into either the splenic or superior mesenteric veins, although some pass directly into the portal vein. The course and distribution of the veins is highly variable even up to the level of the major named vessels.

Short gastric veins

Three to five short gastric veins drain the gastric fundus and the upper part of the greater curvature into the splenic vein or one of its large tributaries.

Left gastroepiploic vein

The left gastroepiploic vein drains both anterior and posterior surfaces of the body of the stomach and the adjacent greater omentum via multiple tributaries. It runs superolaterally along the greater curvature, between the layers of the gastrocolic omentum, and drains into the splenic vein within the gastrosplenic ligament.

In patients requiring replacement of the oesophagus for severe congenital anomalies or after resection for cancer, a gastric tube supplied by the left gastroepiploic artery can be fashioned from the greater curvature of the stomach and anastomosed to the cervical oesophagus ([Arul and Parikh 2008](#)).

The right gastroepiploic artery has been used for coronary artery revascularization in some centres and this may pose a particular hazard if the patient subsequently requires surgery for gastric cancer.

Abnormalities of the intramural vascularity of the stomach are a rare cause of acute upper gastrointestinal haemorrhage. The so-called 'Dieulafoy' lesion occurs most often near the lesser curvature within a few centimetres of the gastro-oesophageal junction and consists of an abnormally large artery that penetrates the muscular coat of the stomach, runs a tortuous course in the submucosa, and protrudes through a small mucosal defect, where it is vulnerable to rupture ([Baxter and Aly 2010](#)).

Right gastroepiploic vein

The right gastroepiploic vein drains the greater omentum, distal body and antrum of the stomach. It runs medially along the greater curvature in the upper part of the gastroduodenal omentum. Just proximal to the pylorus, it passes posteriorly to drain into the superior mesenteric vein below the neck of the pancreas. The right gastroepiploic vein may receive the right colic vein (forming a 'gastroduodenal trunk') and/or superior pancreaticoduodenal vein close to its entry into the superior mesenteric vein (Yamaguchi et al 2002).

Left gastric vein

The left gastric vein drains the upper body and fundus of the stomach. It ascends along the lesser curvature to the oesophageal opening, where it receives several lower oesophageal veins. It then curves posteriorly and medially behind the posterior peritoneal surface of the lesser sac, passing either anterior or posterior to the common hepatic artery. It usually drains into the portal vein at the level of the upper border of the first part of the duodenum, which corresponds to 1–2 cm from the origin of the portal vein (Rebibo et al 2012). In up to one-third of individuals, the left gastric vein terminates in the splenic vein. On rare occasions, it drains into the left portal vein within the liver (Ohkubo 2000), which may be clinically important in portal hypertension.

Right gastric vein

The right gastric vein is typically small and runs along the medial end of the lesser curvature, passing under the peritoneum as it is reflected from the posterior aspect of the pylorus and first part of the duodenum on to the posterior wall of the lesser sac. It drains directly into the portal vein at the level of the first part of the duodenum. It receives the prepyloric vein as it ascends anterior to the pylorus at the level of the pyloric opening. Rarely, the right gastric vein drains directly into a branch of the portal vein within the liver.

Posterior gastric veins

One or more distinct posterior gastric veins may be present, draining the middle of the posterior surface of the stomach into the splenic vein. They may become particularly prominent in portal hypertension (Kimura et al 1990).

Oesophageal and gastric varices

Blood from the oesophageal mucosa normally drains into a submucosal venous plexus, then into a deeper intrinsic venous plexus and finally into peri-oesophageal veins via perforating veins. In the abdominal oesophagus, the perforating veins drain into tributaries of the left gastric vein, whereas, in the lower thoracic oesophagus, they drain into tributaries of the azygos and hemiazygos systems. Bidirectional flow is possible in this region, and accommodates pressure changes that occur during breathing and Valsalva manoeuvres.

Oesophageal and gastric varices are abnormally dilated veins that occur in the submucosal plexus of the distal oesophagus and gastric fundus when portal venous pressure is chronically elevated (typically greater than 15 mmHg). This may develop as a consequence of liver fibrosis or cirrhosis, or portal vein thrombosis, or from a variety of other causes. Portal hypertension leads to the recanalization of occluded embryonic venous channels between venous tributaries of the portal system and the systemic venous circulation (e.g. the ligamentum teres) and progressive dilation of small, naturally occurring venous anastomoses between portal and systemic venous tributaries (Paquet 2000). Valves within these veins become incompetent, permitting retrograde flow and causing the development of varices. Varices in the distal oesophagus are readily visible at endoscopy because they are situated superficially and protrude into the oesophageal lumen (Fig. 64.13); they are vulnerable to rupture and a source of major gastrointestinal bleeding. Gastric varices may also be present on the inferior aspect of the cardiac orifice.

Lymphatic drainage

The stomach has a rich network of lymphatics that connect with lymphatics draining other viscera within the upper abdomen. At the gastro-oesophageal junction, the lymphatics are continuous with those draining the lower oesophagus; in the region of the pylorus, they are continuous with those draining the duodenum and pancreas. In general, lymphatics follow the course of the arteries supplying the stomach. However, many separate groups of nodes are now recognized (Fig. 64.14) and their relationship to the specific regions and vascular territories of the stomach is of great importance during resection of the stomach, particularly for malignancy. Pancreatic and hepatic lymphatics play a significant role in draining the stomach during disease.

Gastrectomy and gastric lymphadenectomy



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INNERVATION

The stomach is innervated by sympathetic and parasympathetic nerves. A variety of neurotransmitters have been identified within pyloric muscle, including acetylcholine, nitric oxide, enkephalins and vasoactive intestinal polypeptide. Inhibition of the sphincter is mediated by nitrergic fibres whilst basal tone is mostly cholinergic (although it should be noted that many other factors, including acid and luminal nutrients, influence pyloric contraction; Ramkumar and Schulze (2005)).

Sympathetic innervation

The sympathetic supply to the stomach originates from the fifth to twelfth thoracic spinal segments, and is mainly distributed via the greater and lesser splanchnic nerves and the coeliac plexus. Peri-arterial plexuses form along the arteries arising from the coeliac trunk. Additional innervation comes from fibres of the hepatic plexus, which pass to the upper body and fundus via the lesser omentum and by direct branches from the greater splanchnic nerves.

Sympathetic activity causes vasoconstriction, inhibits gastric motility and constricts the pylorus. Afferent sensory pathways, including pain, travel with sympathetic efferent nerves.

Parasympathetic innervation

The parasympathetic supply to the stomach is from the vagus nerves (Fig. 64.15). The anterior vagal trunk (formed mainly from fibres from the left vagus nerve within the oesophageal plexus) is often double or even triple, and supplies filaments to the cardiac orifice. The nerve is closely applied to the anterior surface of the outer muscle of the abdominal oesophagus and usually divides near the gastro-oesophageal junction into gastric, pyloric and hepatic branches (Jackson 1949). Upper gastric branches radiate on the anterior surface of the upper body and fundus but the main gastric branch (also known as the anterior nerve of the lesser curvature or greater anterior gastric nerve) lies in the lesser omentum near the lesser curvature. It supplies branches to the body and antrum, and usually terminates near the angular incisure in a crow's foot type of pattern (Shanthi and Sudhayesshayan 2011). The hepatic branch runs almost transversely between the peritoneal layers of the lesser omentum towards its free edge to reach the hilum of the liver, where hepatic branches ramify. From here, some fibres descend adjacent to the hepatic artery to supply the pylorus, duodenum and pancreas. An additional pyloric branch often arises from the greater anterior gastric nerve during its course; this runs inferomedially to the pyloric antrum, where it gives off branches to the pylorus before running superiorly to contribute to the hepatic plexus. Variations in the anterior nerve include accessory pyloric branches and high and low courses of the hepatic and pyloric branches in the lesser omentum.

The posterior vagal trunk usually lies within loose connective tissue immediately posterior and to the right of the oesophagus. It divides into gastric branches and one or more coeliac branches. Gastric branches run behind the cardiac orifice and body of the stomach and extend to the proximal antrum, but do not normally reach the pyloric sphincter. The largest gastric branch (also known as the posterior nerve of the lesser curvature or the greater posterior gastric nerve) descends posteriorly near the lesser curvature. The coeliac branch arises from the posterior vagal trunk and carries the majority of fibres contributing to the coeliac plexus. One or two small hepatic branches may also originate from the coeliac division of the nerve.

The parasympathetic nerve supply is secretomotor to the gastric mucosa and motor to the gastric musculature. It is responsible for coordinated relaxation of the pyloric sphincter during gastric emptying. However, the majority of fibres within the vagus nerves are afferent; these convey gut sensation, including fullness, nausea and probably pain.

Referred pain

Pain sensation from the stomach is poorly localized and, in common with other structures of foregut origin, is referred to the epigastrium. Pain arising from the gastro-oesophageal junction is commonly referred to the lower retrosternal and subxiphoid areas.

MICROSTRUCTURE

The gastric wall consists of the major layers found elsewhere in the gut, i.e. mucosa, submucosa, muscularis externa and serosa, together with





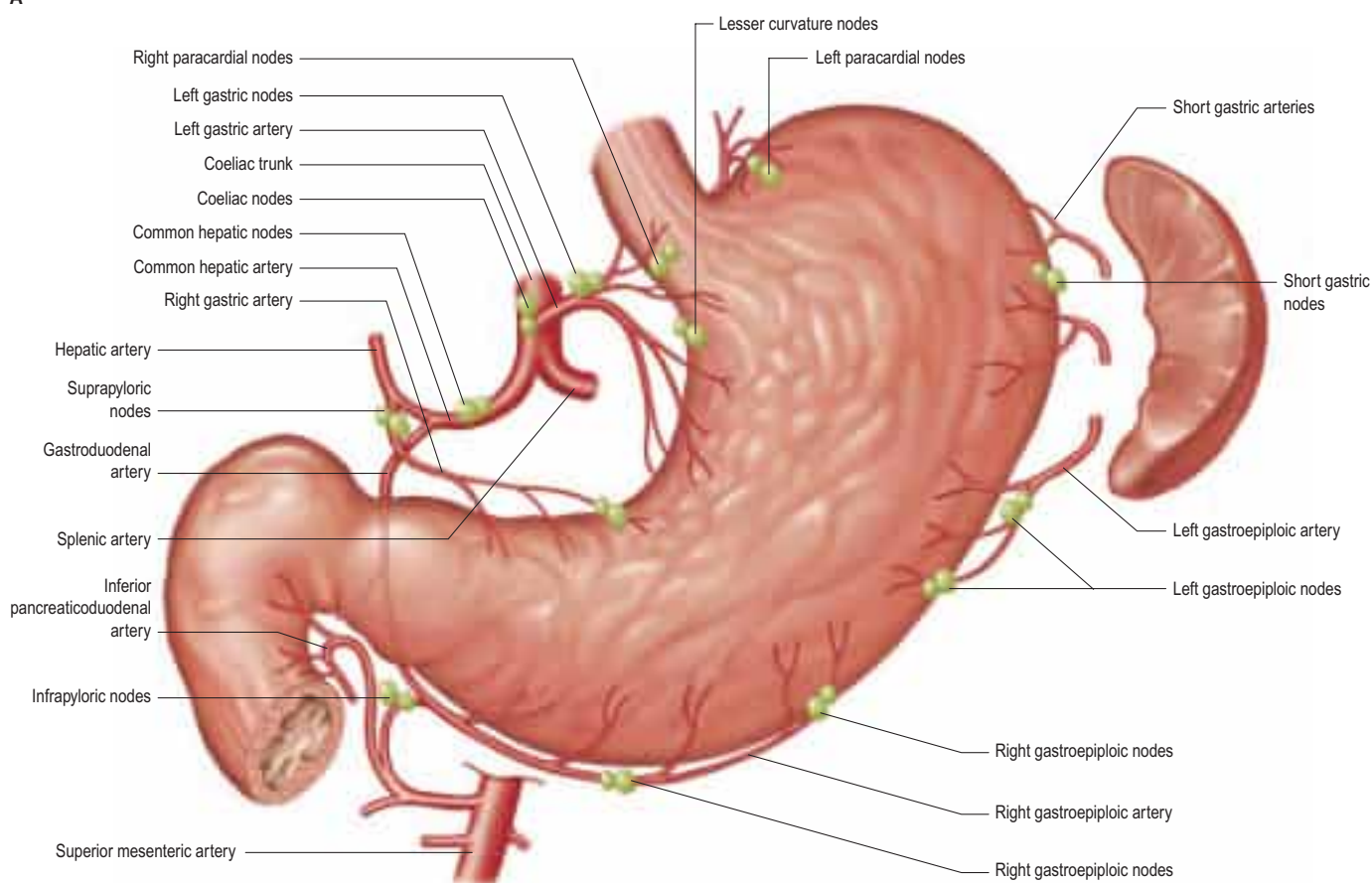
Fig. 64.13 The endoscopic appearance of oesophageal and gastric varices in portal hypertension. **A**, Oesophageal varices. **B**, Fundal varices seen from within the stomach after retroflexion of the gastroscope. **C**, Portal gastropathy (gastric antrum) due to venous congestion of the gastric mucosa. (Courtesy of Professor Mark Stringer.)

Surgery for gastro-oesophageal malignancy involves varying degrees of radical resection of lymph nodes that drain the tumour. Locoregional lymphatic spread can occur in the absence of haematogenous spread and gastric cancer may therefore remain a localized disease, even when lymph nodes are involved (UICC et al 1997). Hence, gastrectomy with meticulous lymphadenectomy can be curative in a proportion of patients with gastric cancer and nodal involvement.

The extent of potential/actual nodal involvement by the tumour is classified as N1 (locoregional nodes specific to the tumour site); N2 (regional and major named vessel nodes draining the tumour); and N3

(wider-draining nodes, including para-aortic nodes). Gastrectomies can be classified according to the node groups excised with the tumour: D1 (removal of the affected portion of the stomach and *en bloc* resection of N1 nodes); D2 (total gastrectomy, including all N1 and N2 nodes); and D3 (total gastrectomy plus extensive lymphadenectomy that includes the associated upper abdominal lymph nodes: namely, pancreatic, superior mesenteric, coeliac, hepatic and transverse colic) (Japanese Gastric Cancer Association 1998, Japanese Research Society for Gastric Cancer 1998).

A



B

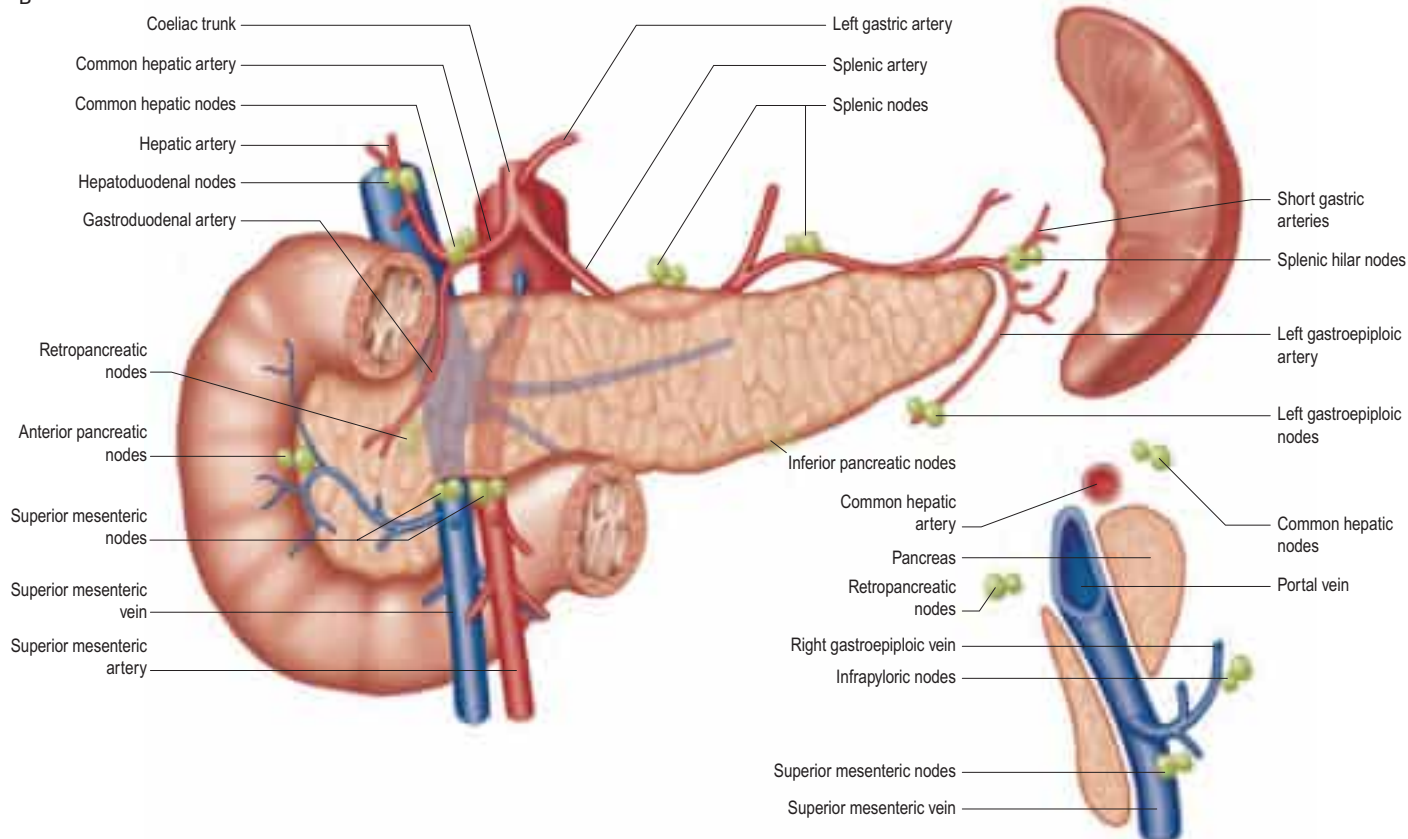


Fig. 64.14 The main lymph node stations of the stomach (A) and upper abdominal viscera (B). The sagittal relationships of the node groups around the neck of the pancreas are shown bottom right. The para-aortic nodes are among the highest nodes for these viscera but have been removed for clarity.

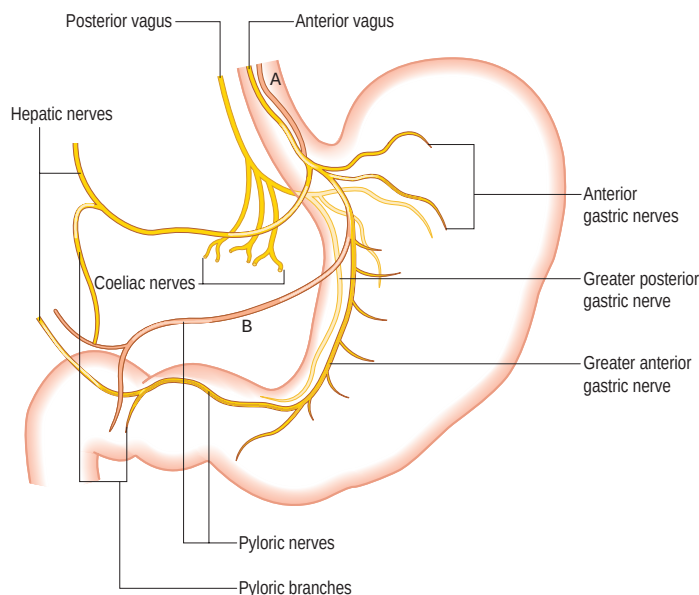


Fig. 64.15 The distribution of the vagal nerves to the stomach. The two most common variations in the anterior vagus are shown in pink. Key: A, multiple main trunks; B, low origin of the hepatic/pyloric branch lying close to the lesser curvature.

gastric vessels and nerves (Figs 64.16–64.17). The microstructure reflects the functions of the stomach as an expandable muscular sac lined by secretory epithelium, although there are local structural and functional variations in this pattern.

Mucosa

The mucosa is a thick layer with a soft, smooth surface that is mostly reddish brown in life but pink in the pyloric region. In the contracted stomach, the mucosa is folded into numerous folds, or rugae, most of which are longitudinal. They are most marked towards the pyloric end and along the greater curvature. The rugae represent large folds in the submucosal connective tissue (see below) rather than variations in the thickness of the mucosa covering them, and they are obliterated when the stomach is distended. As elsewhere in the gut, the mucosa is composed of a surface epithelium, lamina propria and muscularis mucosae.

Epithelium

When viewed microscopically at low magnification, the internal surface of the stomach wall appears honeycombed by small, irregular gastric pits: there are approximately 60 to 100 gastric pits per square millimeter of gastric mucosa, each pit having a diameter of approximately 70 μm and a depth of about 0.2 mm (Lillibridge 1964) (see Figs 64.16–64.17). The base of each gastric pit receives several long, tubular gastric glands that extend deep into the lamina propria as far as the muscularis mucosae. Simple columnar mucus-secreting epithelium covers the entire luminal surface, including the gastric pits, and is composed of a continuous layer of surface mucous cells that release gastric mucus from their apical surfaces to form a thick, protective, lubricant layer over the gastric lining. This epithelium commences abruptly at the Z line at the cardiac orifice, where there is a sudden transition from oesophageal stratified squamous epithelium.

Gastric glands

Although all gastric glands are tubular, they vary in form and cellular composition in different parts of the stomach. They can be divided into three groups: cardiac, principal and pyloric. The most highly specialized are the principal glands.

Principal gastric glands

The principal glands are found in the body and fundus, three to seven opening into each gastric pit. Their junction with the base of the pit is the isthmus; immediately basal to this is the neck; and the remainder is the base. The walls of the gland contain at least five distinct cell types: chief, parietal, mucous neck, stem and neuroendocrine.

Chief (peptic) cells (see Fig. 64.16) are the source of the digestive enzymes pepsin and lipase. They are usually basal in position and

cuboidal in shape, and their nuclei are rounded and euchromatic. They contain secretory zymogen granules and their abundant cytoplasmic RNA renders them strongly basophilic. Parietal (oxyntic) cells are the source of gastric acid and of intrinsic factor, a glycoprotein necessary for the absorption of vitamin B₁₂. They are large, oval and strongly eosinophilic, and have centrally placed nuclei.

Parietal cells occur intermittently along the walls of the more apical half of the gland but can reach as far as the isthmus; they bulge laterally into the surrounding connective tissue. They have a unique ultrastructure related to their ability to secrete hydrochloric acid. The luminal side of the cell is deeply invaginated to form a series of blind-ended channels (canaliculi) that bear numerous irregular microvilli covered by a plasma membrane rich in H⁺/K⁺ ATPase antiporter channels. The latter actively secrete hydrogen ions into the lumen; chloride ions follow along the electrochemical gradient. The mitochondria-rich cytoplasm facing these channels contains a tubulo-vesicular system of abundant fine membranous tubules directed towards the canalicular surface. The precise structure of the cell varies with its secretory phase: when stimulated, the number and surface area of the microvilli increases up to five-fold, probably as a result of the rapid fusion of the tubulo-vesicular system with the plasma membrane. This process is reversed at the end of stimulated secretion, when the excess membrane retreats back into the tubulo-alveolar system and microvilli are lost.

Mucous neck cells are numerous at the necks of the glands and are scattered along the walls of the more basal regions. They are typical mucus-secreting cells, displaying apical secretory vesicles, containing mucins, and basally displaced nuclei; their products are distinct histochemically from those of the superficial mucous cells.

Stem cells are relatively undifferentiated mitotic cells from which the other types of gland cell are derived. They are relatively few in number, and are situated in the isthmus of the gland and the bases of the gastric pits. Stem cells are columnar and possess a few short apical microvilli. They periodically undergo mitosis; their progeny migrate either apically, to differentiate into new surface mucous cells, or basally, to form mucous neck, parietal, chief or neuroendocrine cells. All of these cells have a limited lifespan, especially the mucus-secreting types, and so they are constantly replaced. The typical replacement period for surface mucous cells is 3 days, and that for mucous neck cells is 1 week. Other cell types appear to live much longer.

Neuroendocrine (enteroendocrine) cells occur in all types of gastric gland but more frequently in the body and fundus of the stomach. They are situated mainly in the deeper parts of the glands, among the chief cells. They are basally situated, pleomorphic cells and display irregular nuclei surrounded by granular cytoplasm that contains clusters of large (0.3 μm) secretory granules. Neuroendocrine cells synthesize a number of biogenic amines and polypeptides that are important in the control of gut motility and glandular secretion. They are part of the system of dispersed neuroendocrine cells. In the stomach, they include gastrin-producing G cells; somatostatin-producing D cells; histamine-producing ECL (enterochromaffin-like) cells; serotonin-producing enterochromaffin cells; and ghrelin-producing P/D(1)-type endocrine cells (Sakata and Sakai 2010).

Cardiac glands

Cardiac glands are confined to a small area near the cardiac orifice; some are simple tubular glands, others are compound branched tubular glands. Mucus-secreting cells predominate; parietal and chief cells are present but sparse.

Pyloric glands

Pyloric glands empty via groups of two or three short, convoluted tubes into the bases of the deep gastric pits of the pyloric antrum; the pits occupy about two-thirds of the mucosal depth (Fig. 64.18). The glands are populated mainly by mucus-secreting cells but they also contain neuroendocrine cells, especially G cells, which secrete gastrin when activated by appropriate mechanical stimulation (causing increased gastric motility and secretion of gastric juices). Although parietal and chief cells are scarce, parietal cells are always present in both fetal and postnatal pyloric glands, and may also appear in the duodenal mucosa, proximally near the pylorus, in adult tissue.

Lamina propria

The lamina propria forms a connective tissue framework between the glands. It contains small masses of lymphoid tissue, gastric lymphatic follicles, which resemble solitary intestinal follicles (especially in early life). It also contains a complex periglandular vascular plexus involved in the maintenance of the mucosal environment, e.g. the removal of bicarbonate produced in the tissues as a counterpart to acid secretion, and neural plexuses rich in both sensory and motor terminals.

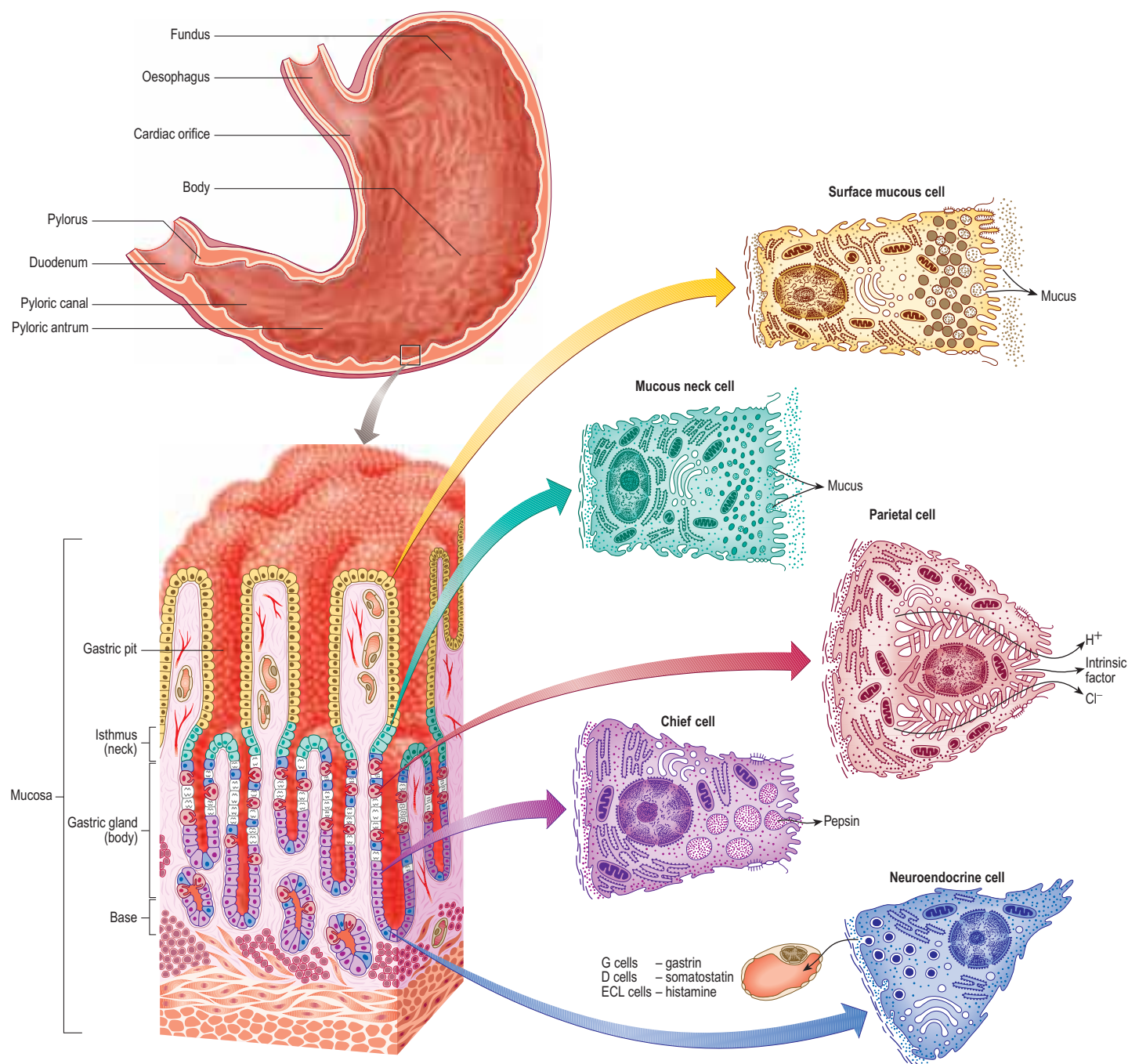


Fig. 64.16 Principal regions of the interior of the stomach and the microstructure of tissues and cells within its wall. Undifferentiated, dividing cells are shown in white. Abbreviations: ECL, enterochromaffin-like.

Muscularis mucosae

The muscularis mucosae is a thin layer of smooth muscle fibres lying external to the layer of glands, arranged as continuous inner circular and outer longitudinal layers, and a discontinuous external circular layer. The inner layer sends strands of smooth muscle cells between the glands; their contraction probably assists in emptying into the gastric pits.

Submucosa

The submucosa is a variable layer of loose connective tissue. It contains thick bundles of collagen, numerous elastin fibres, blood vessels and nervous plexuses, including the ganglionated submucosal (Meissner's) plexus.

Muscularis externa

The muscularis externa is a thick muscle coat immediately under the serosa, with which it is closely connected by subserous loose connective tissue. From innermost outwards, it contains oblique, circular and longitudinal layers of smooth muscle fibres (see Fig. 64.6). The layers are

variably developed in different regions of the stomach and not easily separated. Clasp-like semicircular smooth muscle fibres are present on the right side of the oesophagus and sling-like oblique gastric muscle fibres on the left side; the latter radiate horizontally towards the greater curvature and maintain the angle between the oesophagus and stomach. The circular muscle layer is thicker in the distal pyloric antrum, where it forms the anular pyloric sphincter. The outer longitudinal layer is most pronounced in the upper two-thirds of the stomach.

The muscularis externa enables the stomach to produce churning movements that mix food with the gastric secretions. When the muscles contract, they reduce the volume of the stomach and throw the mucosa into longitudinal folds or rugae (see above). Muscle activity is controlled by a network of unmyelinated autonomic nerve fibres and their ganglia, which lie between the muscle layers in the myenteric (Auerbach's) plexus.

Interstitial cells of Cajal

Like smooth muscle elsewhere in the gut, interstitial cells of Cajal are present within the submucosa and muscular layers of the stomach. These spindle-shaped cells contact nerve fibres and form gap junctions

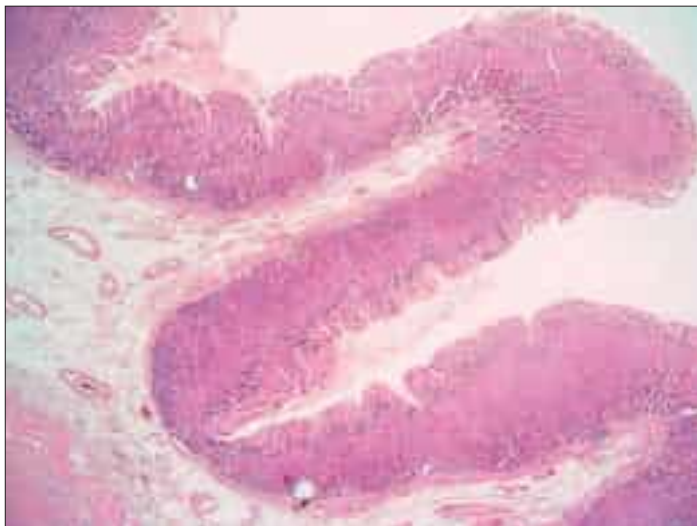


Fig. 64.17 A low-power micrograph showing the stomach wall, thrown into longitudinal folds or rugae that are visible macroscopically. The surface epithelium is infolded microscopically to form gastric pits. Gastric glands extend through the thickness of the mucosal lamina propria and open into the bases of the gastric pits. A muscularis mucosae layer and submucosa follow the contours of the rugae. Part of the external muscularis layers is seen below left. (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

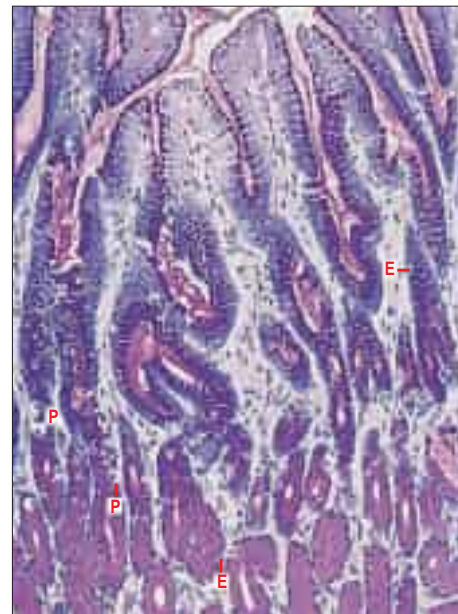


Fig. 64.18 A micrograph showing the pyloric region of the stomach. Pyloric glands are stained with the periodic acid-Schiff (PAS) technique to show mucin (magenta) in the gastric pits and glands. Pale-staining cells are the larger parietal cells (P) and smaller enteroendocrine cells (E). (Courtesy of Dr JB Kerr, Monash University with permission from Kerr JB 1999 Atlas of Functional Histology. London: Mosby.)

with smooth muscle cells, consistent with their role in modulating motor neurotransmission. Interstitial cells of Cajal are involved in the generation of rhythmic gastric slow-wave contractions and play an important role in gastric motility. They have been implicated in the pathogenesis of gastric motility disorders such as diabetic gastric paresis (Negreanu et al 2008) and in the development of gastric stromal tumours (Roggin and Posner 2012).

Serosa or visceral peritoneum

The serosa is an extension of the visceral peritoneum. It covers the entire surface of the stomach other than along the attachments of the greater and lesser omenta to the greater and lesser curvatures, respectively, where the peritoneal layers are separated by vessels and nerves, and over a small posterosuperior area near the cardiac orifice, where the stomach contacts the diaphragm at the reflections of the gastrophrenic and left gastropancreatic folds.

Endoscopic mucosal resection

Endoscopic mucosal resection (EMR) is a technique that can be applied in the oesophagus and stomach in order to excise dysplastic lesions or early cancers that remain confined to the mucosa (Pech et al 2008). EMR is a misnomer since the technique involves resection of both the mucosa and submucosa. The technique is inappropriate for tumours that have invaded the submucosa since these are associated with a significant risk of lymph node metastases and therefore require formal surgical resection (oesophagectomy or gastrectomy) and lymphadenectomy (Ancona 2008).

GASTRIC MOTILITY

In simple terms, the proximal region of the stomach (fundus and upper body) functions mostly as a temporary storage compartment, regulating intragastric pressure and the onward passage of chyme into the distal stomach. Proximal gastric tone decreases with swallowing (receptive relaxation) and in response to gastric distension (gastric accommodation). Its activity is modulated by enterogastric reflexes (e.g. acid, protein or fat in the duodenum inhibits proximal gastric muscle contraction) and gut hormones (e.g. cholecystokinin, gastrin, glucagon and vasoactive intestinal polypeptide). The vagus nerve plays a key role in these reflexes. In contrast to this activity, the muscle of the distal region of the stomach exhibits strong phasic contractions that increase in amplitude towards the pylorus; these occur about three times per minute and act to grind the food mechanically and propel it towards the pylorus. Solid material lags behind liquids, and when the contents of the antrum reach the contracted pylorus, only the liquid and small particulate solids can be expelled into the duodenum; larger solid matter is retropulsed back into the stomach for further breakdown (Patrick and Epstein 2008).

Bonus e-book images

Fig. 64.13 The endoscopic appearance of oesophageal and gastric varices in portal hypertension.

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